DATASHEET

DDR2/3/3L/4/LPDDR2/3/4/4X Combo Controller

Version: 2.1

September 2024

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Revision History

Version	Date	Description
2.1	September 2024	Updated interface description
2.0	March 2024	New version
1.0	August 2020	Initial Release

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1

Introduction

The INNO DDR2/3/3L/4/LPDDR2/3/4/4X Controller (IDC) combined with INNO DDR2/3/3L/4/LPDDR2/3/4/4X PHY is a complete memory interface solution for DDR2/3/3L/4/LPDDR2/3/4/4X memory systems. This complete mixed-signal IP solution is designed to provide DDR2/3/3L/4/LPDDR2/3/4/4X SDRAM connectivity in a high bandwidth and low latency system targeted to a specific fabrication process.

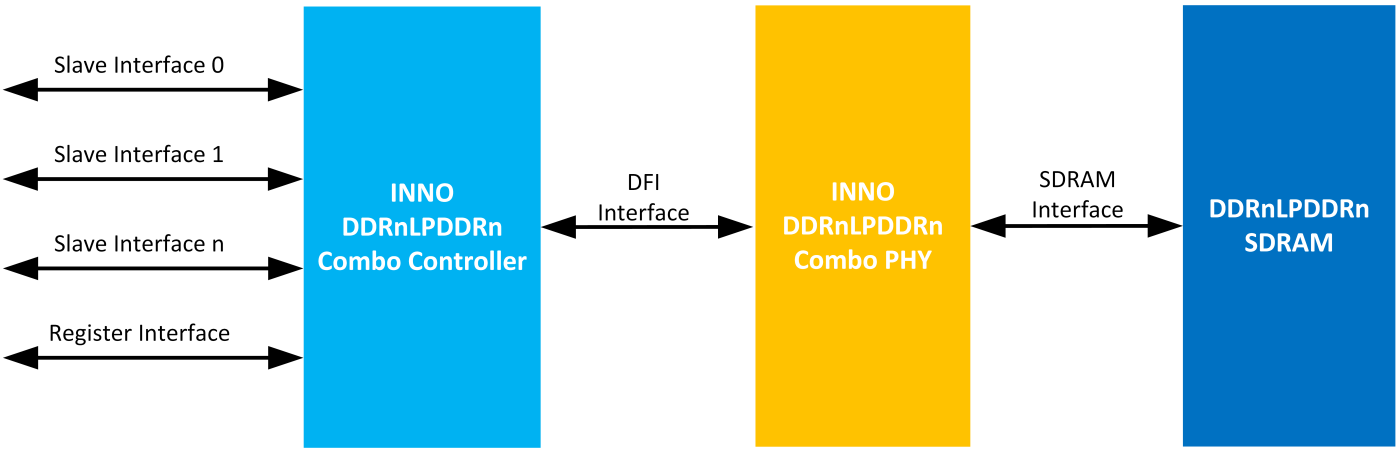

Figure 1. System-Level Block Diagram

The IDC can support up to 8 slave memory access ports from the host side. Each slave port can be connected to the IDC through the standard AMBA AXI bus interfaces. The register interface is connected to the IDC through the standard AMBA APB bus interfaces.

The Controller receives AXI transactions that are queued internally and scheduled to access the SDRAM according to the SDRAM protocol timing requirements.

The Controller issues commands compliant with the DFI protocol through the DFI interface to the PHY module, which sends command/data to the SDRAM and receives data from the SDRAM.

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1.1. Features

1.1.1. General Features

Feature		DDR			LPDDR		
		2	3/3L	4	2	3	4/4X
At Speed DFT		√	√	√	√	√	√
FREQ Ratio	core_clk:CK:Data_Rate = 1:2:4	√	√	√	√	√	√
RANK	1	√	√	√	√	√	√
	2	√	√	√	√	√	√
	4	√	√	√	√	√	√
DQ Width	Full	√	√	√	√	√	√
	Half	√	√	√	√	√	√
	Quarter (DQ ≥ 32)	√	√	√	√	√	√
CAM Depth Configurable		√	√	√	√	√	√
2T Timing Function		**X**	√	√	**X**	**X**	**X**
2N Timing Function		**X**	**X**	**X**	**X**	**X**	**X**
Gear Down Function		**X**	**X**	**X**	**X**	**X**	**X**
Self-Refresh	Auto Self-Refresh Enter or Exit When Lack of Transaction	√	√	√	√	√	√
	Register Self-Refresh Enter or Exit	√	√	√	√	√	√
	HWFFC Self-Refresh Enter or Exit	√	√	√	√	√	√
	Partial-array Self-Refresh (PASR)	**X**	**X**	**X**	**X**	**X**	**X**
Power Down	Auto Power Down Enter or Exit When Lack of Transaction	√	√	√	√	√	√
	Register Power Down Enter or Exit	√	√	√	√	√	√
Auto Refresh	Auto Refresh All Bank Single/Burst	√	√	√	√	√	√
	Auto Refresh Per-Bank Single/Burst	√	√	√	√	√	√
	Auto Refresh Disable	√	√	√	√	√	√
	Refresh Management	**X**	**X**	**X**	**X**	**X**	**X**
	Temperature-Controlled Refresh and Timing Parameter (Derate)	**X**	**X**	**X**	**X**	**X**	√

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Feature		DDR			LPDDR		
		2	3/3L	4	2	3	4/4X
	Per-bank Refresh for Concurrent Bank Operation	X	X	X	X	X	√
Arbiter	Page-match Limit	√	√	√	√	√	√
	Round Robin Based on Configurable Priority	√	√	√	√	√	√
	QoS	√	√	√	√	√	√
Page Management	Keep Page Open after the Last Access	√	√	√	√	√	√
	Close Page after the Last Access When the Configured Time Reaches	√	√	√	√	√	√
	Close Page after the Last Access Immediately (Auto Precharge)	√	√	√	√	√	√
Write Combine		√	√	√	√	√	√
Coherency		X	X	X	X	X	X
Data Mask		√	√	√	√	√	√
Data Bus Inversion		X	X	√	X	X	√
Data Mask Inversion Mode		X	X	X	X	X	√
Read Modify Write		√	√	√	√	√	√
OSC Function		X	X	X	X	X	X
Mode Register Access	Write	√	√	√	√	√	√
	Read	√	√	√	√	√	√
Post Package Repair		X	X	X	X	X	X
Physical Address Map Configurable	Normal Mode	√	√	√	√	√	√
	Hash Mode	√	√	√	√	√	√
Sideband ECC	Scrub Function	√	√	√	√	√	√
	Periodic inspection	√	√	√	√	√	√
Inline ECC	Scrub Function	√	√	√	√	√	√
	Periodic inspection	√	√	√	√	√	√
Data Scramable		X	X	X	X	X	X
DFI Interface Timing Control	Auto Calculate	√	√	√	√	√	√
	Bypass Mode	√	√	√	√	√	√

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Feature		DDR			LPDDR		
		2	3/3L	4	2	3	4/4X
Dynamic Frequency Change	Frequency Point 1	√	√	√	√	√	√
	Frequency Point 2	√	√	√	√	√	√
	Frequency Point 4	**X**	**X**	**X**	**X**	**X**	√
Programmable and On-The-Fly Burst Lengths	BL = 8	√	√	√	√	√	**X**
	BL = 16	**X**	**X**	**X**	**X**	**X**	√
ZQ Calibration	Command-based Mode	√	√	√	√	√	√
	Background Mode	**X**	**X**	**X**	**X**	**X**	**X**
Enhance Mode		√	√	√	√	√	√
Module Level Clock Gate		√	√	√	√	√	√
Programmable Read and Write Latencies (RL/WL)		√	√	√	√	√	√
Memory BIST Function		√	√	√	√	√	√
PHY BIST Function		√	√	√	√	√	√
Auto Clock Gating for Controller Clock		√	√	√	√	√	√
Hardware Fast Frequency Change		√	√	√	√	√	√
IO Retention		√	√	√	√	√	√
Efficiency and Command Monitor (DFX)		√	√	√	√	√	√
Interrupt		√	√	√	√	√	√
Multi-purpose Register		**X**	√	√	√	√	√
Max Speed*	1066Mbps	√	√	√	√	√	√
	2133Mbps	**X**	√	√	**X**	√	√
	3200Mbps	**X**	**X**	√	**X**	**X**	√
	4266Mbps	**X**	**X**	**X**	**X**	**X**	√

Note: *In theory, the Innosilicon Controller can support the maximum speed specified by the protocol for each SDRAM. However, the actual maximum supported speed may be influenced by factors such as the process and design requirements.*

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1.1.2. DFI Features

Feature	DDR			LPDDR		
	2	3/3L	4	2	3	4/4X
DFI 3.1	√	√	√	√	√	√
DFI 4.0	√	√	√	√	√	√
DFI 5.0	**X**	**X**	**X**	**X**	**X**	**X**
DFI Command Interface	√	√	√	√	√	√
DFI Data Interface	√	√	√	√	√	√
DFI Controller Update Interface	√	√	√	√	√	√
DFI PHY Master Interface	√	√	√	√	√	√
DFI Low Power Interface	√	√	√	√	√	√
DFI Error Response Interface	**X**	**X**	**X**	**X**	**X**	**X**
DFI Controller Training Interface	**X**	**X**	**X**	**X**	**X**	**X**
DFI Disconnect Interface	**X**	**X**	**X**	**X**	**X**	**X**

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1.1.3. AXI Features

Feature	DDR			LPDDR		
	2	3/3L	4	2	3	4/4X
AXI 3.0	√	√	√	√	√	√
AXI 4.0	√	√	√	√	√	√
AXI Port Number Configurable	√	√	√	√	√	√
AXI Clock Synchronous to the Core Clock	√	√	√	√	√	√
AXI Clock Asynchronous to the Core Clock	√	√	√	√	√	√
AXI QoS Feature (Each port QoS value configurable)	√	√	√	√	√	√
AXI Two Command Queue Feature (HPR and LPR/VPR)	√	√	√	√	√	√
AXI Exclusive Access (Max to 8 monitors for all AXI ports)	√	√	√	√	√	√
AXI Wrap Burst	√	√	√	√	√	√
AXI Increment Burst	√	√	√	√	√	√
AXI Fix Burst	**X**	**X**	**X**	**X**	**X**	**X**
AXI Outstanding Configurable	√	√	√	√	√	√
AXI Narrow Transition Support	√	√	√	√	√	√
AXI Unaligned Transition Support	√	√	√	√	√	√
Auto Clock Gating for AXI Clock of Each Port	√	√	√	√	√	√
AXI Low Power Interface	√	√	√	√	√	√
AXI Write Interleave	**X**	**X**	**X**	**X**	**X**	**X**
AXI Read Data Out of Order for Per Port	√	√	√	√	√	√
AXI Data Width Mix (64/128/256)	√	√	√	√	√	√
AXI ID Width Configurable	√	√	√	√	√	√
AXI Upsize Support	√	√	√	√	√	√
AXI Down Size Support	√	√	√	√	√	√
AXI Order Keep Configurable	√	√	√	√	√	√

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Feature	DDR			LPDDR		
	2	3/3L	4	2	3	4/4X
AXI Address Exceed Range Interrupt	√	√	√	√	√	√
AXI 4K Boundary	√	√	√	√	√	√
AXI Cache Interface	X	X	X	X	X	X
AXI User Interface	X	X	X	X	X	X
AXI Proctect Interface	X	X	X	X	X	X
AXI Urgent Interface	X	X	X	X	X	X

1.1.4. Register Access Interface Features

Feature	DDR			LPDDR		
	2	3/3L	4	2	3	4/4X
APB 2.0	X	X	X	X	X	X
APB 3.0 (Default)	√	√	√	√	√	√
APB 4.0 (Optional)	√	√	√	√	√	√

1.2. Standards Compliance

The controller supports the latest version of specifications:

- JESD209-2B (LPDDR2)
- JESD209-3 (LPDDR3)
- JESD209-4D (LPDDR4/4X)
- JESD79-2C (DDR2)
- JESD79-3 (DDR3/3L)
- JESD79-4A (DDR4)

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1.3. Performance

1.3.1. Gate Count

Table 1. Gate Count for DDR2/3/3L/4/LPDDR2/3/4/4X Combo Controller

Frequency	Cell Area (DC Result)
800MHz	1500K gates

1.3.2. Latency

Table 2. Latency of the IDC

Parameter	From	To	Cycle	Cycle (Sync)
WR	Wcmd on AXI Interface	Wcmd on the CMD Queue (AXI clock domain)	5 * T_axi_clk	5 * T_axi_clk
	Wcmd on the CMD Queue (AXI clock domain)	Wcmd on the core clock domain	1 * T_axi_clk	5 * T_axi_clk
	Wcmd on the core clock domain	Wcmd on the CMD Queue (Core clock domain)	1 * T_core_clk	1 * T_core_clk
	Wcmd on the CMD Queue (Core clock domain)	Wcmd on the DFI interface	4 * T_core_clk	4 * T_core_clk
	Wcmd on the DFI interface	Wcmd on the SDRAM	11 * T_core_clk	11 * T_core_clk
RD	Rcmd on AXI Interface	Rcmd on the CMD Queue (AXI clock domain)	5 * T_axi_clk	5 * T_axi_clk
	Rcmd on the CMD Queue	Rcmd on the core clock domain	6 * T_axi_clk	1* T_axi_clk

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Table 2. Latency of the IDC (continued)

Parameter	From	To	Cycle	Cycle (Sync)
	(AXI clock domain)			
	Rcmd on the core clock domain	Rcmd on the CMD Queue (Core clock domain)	1 * T_core_clk	1 * T_core_clk
	Rcmd on the CMD Queue (core clock domain)	Rcmd on the DFI interface	2 * T_core_clk	2 * T_core_clk
	Rcmd on the DFI interface	Rcmd on the SDRAM	4 * T_core_clk	4 * T_core_clk
	Rcmd on the SDRAM	Rdata on the DFI interface	28 * T_core_clk	28 * T_core_clk
	Rdata on the DFI interface	Rdata on the Rdata queue (Core clock domain)	11 * T_core_clk	11 * T_core_clk
	Rdata on the Rdata queue (Core clock domain)	Rdata on the Rdata queue (AXI clock domain)	5 * T_core_clk	1 * T_core_clk
	Rdata on the Rdata queue (AXI clock domain)	Rcmd on AXI Interface	4 * T_axi_clk	4 * T_axi_clk

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2

Architecture

2.1. System Architecture

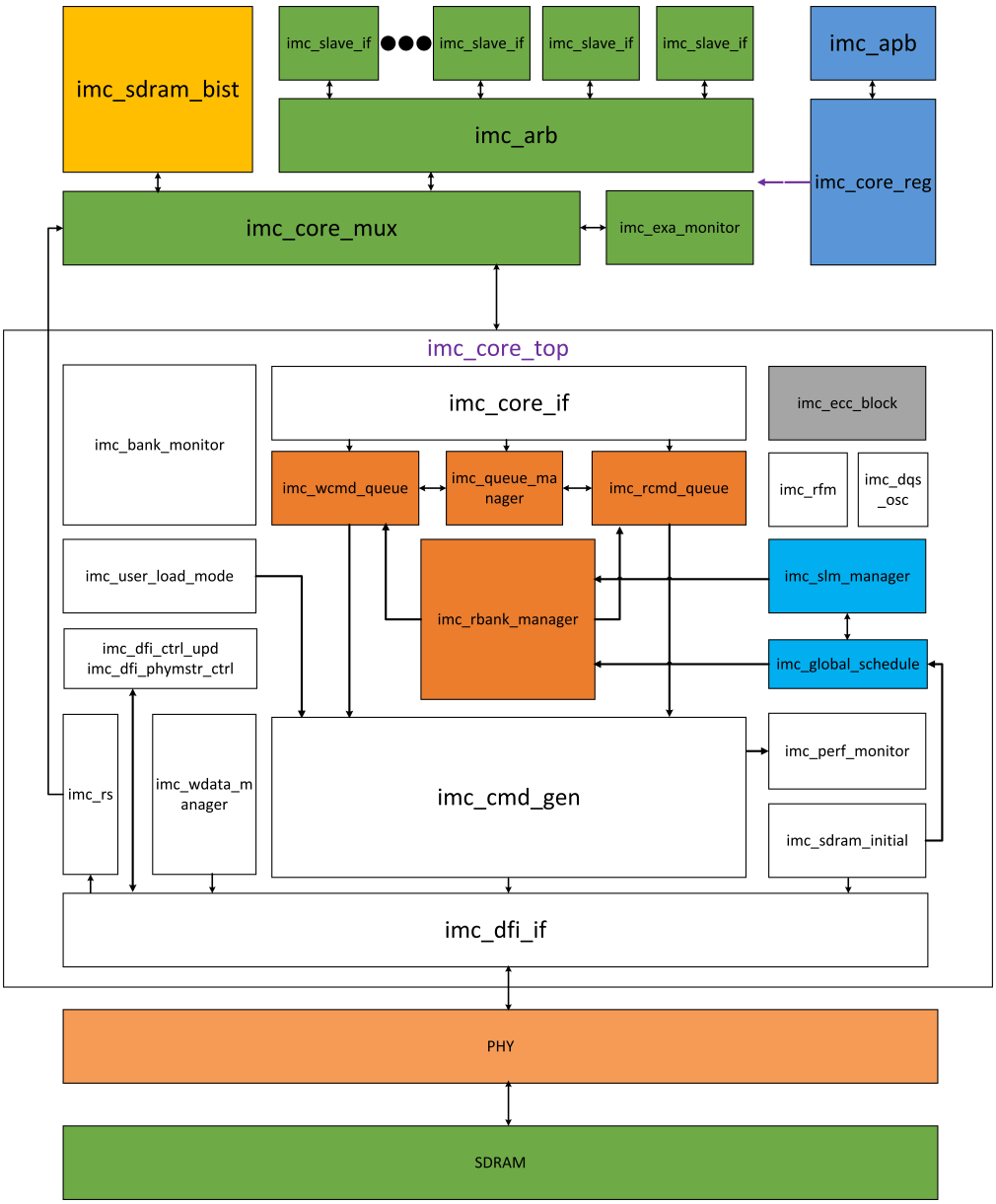

Figure 2. DDR2/3/3L/4/LPDDR2/3/4/4X Combo Controller

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2.2. Block Description

The IDC includes the following components:

- **Slave Interface:** provides the interface to the application level. It provides interface protocol handling, data/command buffering, reordering for read data and address alignment.
- **Arbiter:** provides priority based on arbitration between the requests issued by the slave interfaces.
- **Controller Core:** contains multiple command queues used to store the aligned commands from the slave interface. The controller optimizes the schedule of commands based on the First Ready–First Coming–First Service algorithm.
- **Register Interface Control:** provides interface protocol handling of the register interface.

2.3. Clock and Reset Structure

The signal **core_clk** is the clock of the core which has the same frequency with the PHY clock (**dfi_clk1x**). The signal **core_rstn** is the reset signal of the core. The controller will add the asynchronized reset and synchronized release module to the **core_rstn** and generate **core_rstn_sync**.

The signal **aclk_*** is the clock of the AXI Slave module. Each port of the AXI Slave has an independent clock without data exchange. The signal **aresetn_*** is the reset signal of the AXI Slave. The controller will add the asynchronized release and synchronized release module to the **aresetn_*** and generate **aresetn_sync**.

The signal **apb_pclk** is the clock of the APB Slave and the signal **apb_presetn** is the reset signal of the APB Slave. The controller will add the asynchronized reset and synchronized release module to the **apb_presetn** and generate **presetn_sync**.

The **core_clk**, **aclk_*** and **apb_pclk** are independent asynchronized clock.

In the default mode, the controller does not incorporate the scan mux into the clock path. This design choice allows users to utilize the OCC module for conducting high-speed testing.

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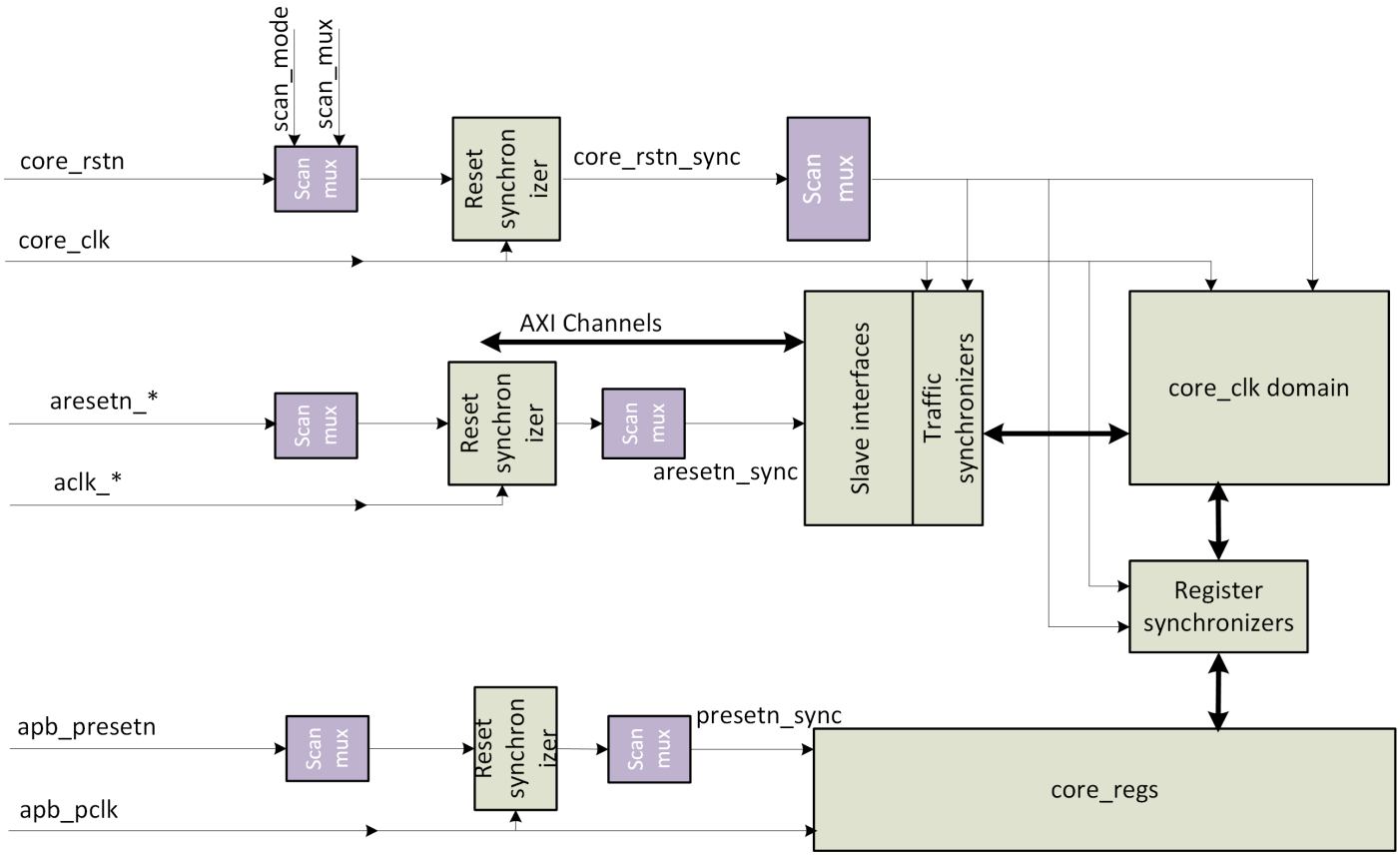

Figure 3. Clock and Reset Structure for Default Mode

If the user opts not to utilize the OCC module in the top-level design but still intends to introduce a scan mux for the clock path, the user should include "**`+define + OCC_DIS`**" directive, which serves to augment the scan logic for the clock path accordingly.

The clock and reset structure is shown in the following figure. The design has already integrated the scan mux cell, and both the clock mux and reset mux components are prepared for DFT purposes.

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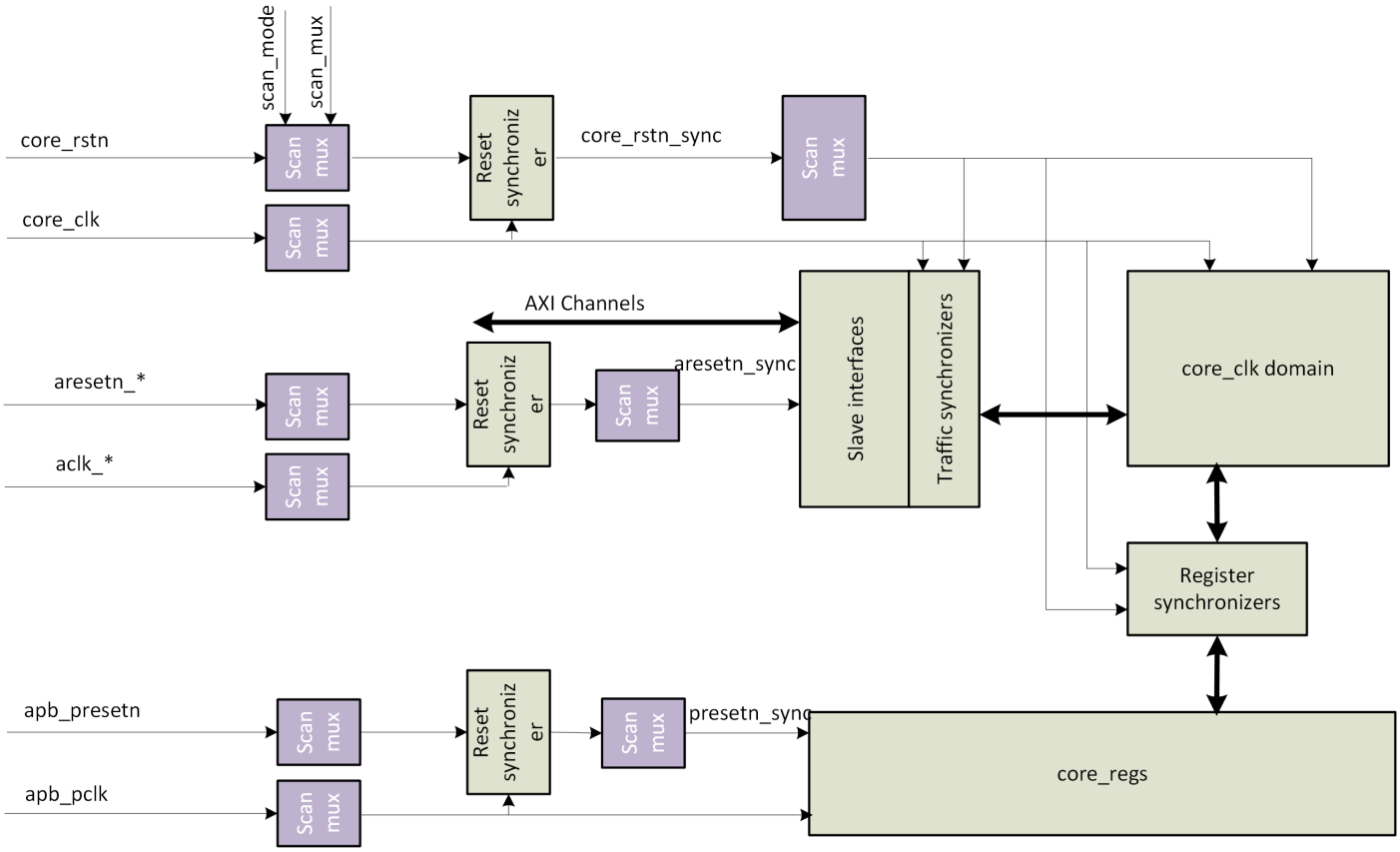

Figure 4. Clock and Reset Structure with + Define + OCC_DIS

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3

Signal Descriptions

3.1. I/O Diagram

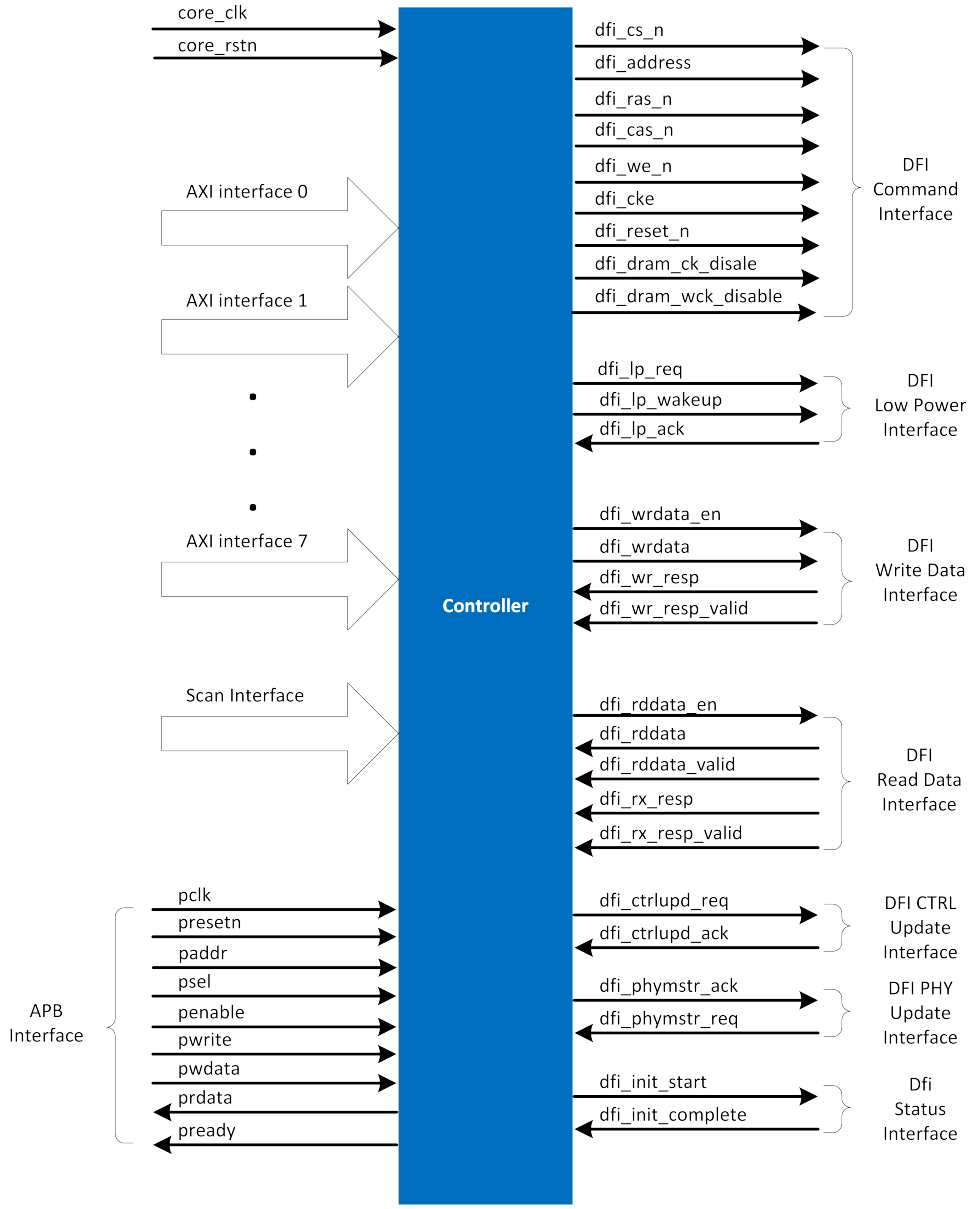

Figure 5. Top-level I/O Diagram

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3.2. AXI Interface

Table 3. AXI Interface Signals

Signal Name	I/O	Description
AXI Clock and Reset		
aclk_*	I	AXI interface clock signal. All signals are sampled on the rising of this signal. Clock for each port can be different.
aresetn_*	I	AXI interface reset signal. **Active State**: Low **Synchronous to**: aclk_*
AXI Write Address Channel		
awid_*[7:0]	I	Write address ID. This signal is the identification tag for the write address group of signals. The width can be changed through IP parameters. **Synchronous to**: aclk_*
awaddr_*[34:0]	I	Write address. Gives the address of the first transfer in a write burst transaction. The width is determined by the maximum capacity that needs to be accessed and can be changed through the IP parameters. **Synchronous to**: aclk_*
awlen_*[7:0]	I	Burst length. Gives the exact number of transfers in a burst. **Synchronous to**: aclk_*
awsize_*[2:0]	I	Burst size. Indicates the size of each transfer in the burst. **Synchronous to**: aclk_*
awburst_*[1:0]	I	Burst type.

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Table 3. AXI Interface Signals (continued)

Signal Name	I/O	Description
		The burst type, coupled with the size information, details how the address for each transfer within the burst is calculated. **Synchronous to**: aclk_*
awlock_*[1:0]	I	Lock type. Provides additional information about the atomic characteristics of the transfer. For AXI 4.0, **awlock_*[1]** should be tied to 0. **Active State**: High **Synchronous to**: aclk_*
awcache_*[3:0]	I	Cache type. Indicates the bufferable, cacheable, write-through, write-back and allocates attributes of the transaction. **Synchronous to**: aclk_*
awprot_*[2:0]	I	Protection type. Indicates the normal, privileged or secure protection level of the transaction and whether the transaction is a data access or an instruction access. **Synchronous to**: aclk_*
awqos_*[3:0]	I	The QoS identifier sent for each write transaction. Implemented only in AXI4. **Synchronous to**:aclk_*
awvalid_*	I	Write address valid. Indicates that valid write address and control information are available. **Active State**: High

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Table 3. AXI Interface Signals (continued)

Signal Name	I/O	Description
		Synchronous to: aclk_*
awready_*	O	Write address ready. Indicates that the slave is ready to accept an address sand associated control signals. **Active State**: High **Synchronous to**: aclk_*
AXI Write Data Channel		
wid_*[7:0]	I	Write ID tag. This signal is the ID tag of the write data transfer. The width can be changed through the IP parameters. This signal is removed from AXI4.0 protocol. **Synchronous to**: aclk_*
wdata_*[2^n-1:0]	I	Write data. The write data bus can be 2^3, 2^4, 2^5, ..., 2^n bits wide. e.g.: If the n equals 7, then the write data bus can be 8, 16, 32, 64, and 128 bits wide. **Synchronous to**: aclk_* ***Note:*** *2^n represents the bit width of the AXI bus.*
wstrb_*[2^n/8-1:0]	I	Write strobe. Indicates which byte lanes to update in memory. **Synchronous to**: aclk_* ***Note:*** *2^n represents the bit width of the AXI bus.*
wlast_*	I	Write last.

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Table 3. AXI Interface Signals (continued)

Signal Name	I/O	Description
		Indicates the last transfer in a write burst. **Active State**: High **Synchronous to**: aclk_*
wvalid_*	I	Write valid. Indicates that valid write data and strobes are available. **Active State**: High **Synchronous to**: aclk_*
wready_*	O	Write ready. Indicates that the slave can accept the write data. **Active State**: High **Synchronous to**: aclk_*
AXI Write Response Channel		
bid_*[7:0]	I	Write response ID. The identification tag of the write response. The width can be changed through the IP parameters. **Synchronous to**: aclk_*
bresp_*[1:0]	I	Write response. Indicates the status of the write transaction. **Synchronous to**: aclk_*
bvalid_*	O	Write response valid. Indicates that a valid write response is available. **Active State**: High **Synchronous to**: aclk_*

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Table 3. AXI Interface Signals (continued)

Signal Name	I/O	Description
bready_*	I	Response ready. Indicates that the master can accept the response information. **Active State**: High **Synchronous to**: aclk_*
AXI Read Address Channel		
arid_*[7:0]	I	Read address ID. This signal is the identification tag for the read address group of signals. The width can be changed through the IP parameters. **Synchronous to**: aclk_*
araddr_*[34:0]	I	Read address. Gives the initial address of a read burst transaction. The width is determined by the maximum capacity that needs to be accessed and can be changed through the IP parameters. **Synchronous to**: aclk_*
arlen_*[7:0]	I	Burst length. Gives the exact number of transfer in a burst. **Synchronous to**: aclk_*
arsize_*[2:0]	I	Burst size. Indicates the size of each transfer in the burst. **Synchronous to**: aclk_*
arburst_*[1:0]	I	Burst type. This signal couples with the size information, details how the address for each transfer is calculated.

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Table 3. AXI Interface Signals (continued)

Signal Name	I/O	Description
		Synchronous to: aclk_*
arlock_*[1:0]	I	Lock type. Provides additional information about the atomic characteristics of transfer. For AXI 4.0, **arlock_*[1]** should be tied to 0. **Active State**: High **Synchronous to**: aclk_*
arcache_*[3:0]	I	Cache type. Provides additional information about the cacheable characteristics of transfer. **Synchronous to**: aclk_*
arprot_*[2:0]	I	Protection type. Provides protection unit information for the transaction. **Synchronous to**: aclk_*
arqos_*[3:0]	I	QoS identifier sent for each read transaction. Implemented only in AXI 4.0. **Synchronous to**: aclk_*
arvalid_*	I	Read address valid. When HIGH, indicates that the read address and control information is valid and will remain stable until the address acknowledge signal. **Active State**: High **Synchronous to**: aclk_*
arready_*	O	Read address ready.

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Table 3. AXI Interface Signals (continued)

Signal Name	I/O	Description
		Indicates that the slave is ready to accept an address and associated control signals. **Active State**: High **Synchronous to**: aclk_*
AXI Read Data Channel		
rid_*[7:0]	O	Read ID tag. This signal is the ID tag of the read data group of signals. The width can be changed through the IP parameters. **Synchronous to**: aclk_*
rdata_*[2^n-1:0]	I	Read data. The read data bus can be 2^3, 2^4, 2^5, ..., 2^n bits wide. e.g.: If the n equals 7, then the read data bus can be 8, 16, 32, 64, and 128 bits wide. **Synchronous to**: aclk_* ***Note:*** *2^n represents the bit width of the AXI bus.*
rresp_*[1:0]	O	Read response. Indicates the status of the read transfer. **Synchronous to**: aclk_*
rlast_*	O	Read last. Indicates the last transfer in a read burst. **Active State**: High **Synchronous to**: aclk_*
rvalid_*	O	Read valid.

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Table 3. AXI Interface Signals (continued)

Signal Name	I/O	Description
		Indicates that the required data is available and the read transfer can complete. **Active State**: High **Synchronous to**: aclk_*
rready_*	I	Read ready. Indicates that the master can accept the read data and response information. **Active State**: High **Synchronous to**: aclk_*
AXI Low Power Interface		
csysreq_*	I	System low-power request. This signal is a request from the system clock controller for the periperal to enter a low-power state. **Active State**: Low **Synchronous to**: aclk_*
csysack_*	O	Low-power request acknowledgement This signal is the acknowledgement from a peripheral of a system low-power request. **Active State**: Low **Synchronous to**: aclk_*
cactive_*	O	Clock active. This signal indicates that the peripheral requires its clock signal. **Active State**: High **Synchronous to**: aclk_*

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3.3. DFI Interface

Table 4. DFI Interface Signals

Signal Name	I/O	Description
DFI Clock and Reset Signals		
core_clk	I	The clock of the IDC's core. Clock from system. It is the same frequency with the PHY DFI Interface.
core_rstn	I	DFI Clock Domain Reset System reset synchronized to core_clk. **Active State**: Low **Synchronous to**: core_clk
DFI Command Interface		
dfi_address[39:0]	O	DFI address bus. These signals define the address information that is intended for the DRAM memory devices for all control commands. **For DDR4:** [14] is mapped to WE_n for P0. [34] is mapped to WE_n for P1. [15] is mapped to CAS_n for P0. [35] is mapped to CAS_n for P1. [16] is mapped to RAS_n for P0. [36] is mapped to RAS_n for P1. **For LPDDR23:** [9:0] for P0 CMD0. [19:10] for P1 CMD0. [29:20] for P0 CMD1.

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Table 4. DFI Interface Signals (continued)

Signal Name	I/O	Description
		[39:30] for P1 CMD1. **For LPDDR4/4X:** [9:0] for the P0. [19:10] for the P1. [29:20] for the P2. [39:30] for the P3.
dfi_cs_n[2*n*-1:0]	O	DFI chip select bus. These signals define the CS information that is intended for the DRAM devices for all control commands. The bit width of this signal is determined by the number of ranks. Taking a 2-rank and 4-rank design as an example, the mapped relationship is as follow: **For DDR23LPDDR23:** [0] is mapped to RANK0 for P0. [1] is mapped to RANK1 for P0. [2] is mapped to RANK0 for P1. [3] is mapped to RANK1 for P1. **For DDR4:** [0] is mapped to RANK0 for P0. [1] is mapped to RANK1 for P0. [2] is mapped to RANK0 for P1. [3] is mapped to RANK1 for P1. [4] is mapped to RANK2 for P0. [5] is mapped to RANK3 for P0. [6] is mapped to RANK2 for P1. [7] is mapped to RANK3 for P1.

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Table 4. DFI Interface Signals (continued)

Signal Name	I/O	Description
		For LPDDR4/4X: [0] is mapped to RANK0 for P0. [1] is mapped to RANK1 for P0. [2] is mapped to RANK0 for P1. [3] is mapped to RANK1 for P1. [4] is mapped to RANK0 for P2. [5] is mapped to RANK1 for P2. [6] is mapped to RANK0 for P3. [7] is mapped to RANK1 for P3. **Voltage Range**: 0 V–VDD **Active State**: Low **Synchronous to**: core_clk **Registered**: Yes ***Note:*** *n represents the number of ranks.*
dfi_ras_n[1:0]	O	DFI row address strobe bus. These signals define the RAS information that is intended for the DRAM devices for all control commands. This signal is only for DDR23. **Voltage Range**: 0 V–VDD **Active State**: Low **Synchronous to**: core_clk **Registered**: Yes
dfi_cas_n[1:0]	O	DFI column address strobe bus. These signals define the CAS information that is intended for the DRAM devices for all control commands.

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Table 4. DFI Interface Signals (continued)

Signal Name	I/O	Description
		This signal is only for DDR23. **Voltage Range**: 0 V–VDD **Active State**: Low **Synchronous to**: core_clk **Registered**: Yes
dfi_we_n[1:0]	O	DFI write enable bus. These signals define the WEN information that is intended for the DRAM devices for all control commands. This signal is only for DDR23. **Voltage Range**: 0 V–VDD **Active State**: Low **Synchronous to**: core_clk **Registered**: Yes
dfi_cke[2*n*-1:0]	O	DFI clock enable bus. These signals define the CKE information that is intended for the DRAM devices for all control commands. The bit width of this signal is determined by the number of ranks. Taking a 2-rank and 4-rank design as an example, the mapped relationship is as follow: **For DDR23LPDDR23:** [0] is mapped to RANK0 for P0. [1] is mapped to RANK1 for P0. [2] is mapped to RANK0 for P1. [3] is mapped to RANK1 for P1. **For DDR4:** [0] is mapped to RANK0 for P0.

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Table 4. DFI Interface Signals (continued)

Signal Name	I/O	Description
		[1] is mapped to RANK1 for P0. [2] is mapped to RANK0 for P1. [3] is mapped to RANK1 for P1. [4] is mapped to RANK2 for P0. [5] is mapped to RANK3 for P0. [6] is mapped to RANK2 for P1. [7] is mapped to RANK3 for P1. **For LPDDR4/4X:** [0] is mapped to RANK0 for P0. [1] is mapped to RANK1 for P0. [2] is mapped to RANK0 for P1. [3] is mapped to RANK1 for P1. [4] is mapped to RANK0 for P2. [5] is mapped to RANK1 for P2. [6] is mapped to RANK0 for P3. [7] is mapped to RANK1 for P3. **Voltage Range**: 0 V–VDD **Active State**: High **Synchronous to**: core_clk **Registered**: Yes ***Note:*** *The n represents the number of ranks.*
dfi_odt[2*n*-1:0]	O	DFI on-die termination control bus. These signals define the ODT information that is intended for the DRAM devices for all control commands.

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Table 4. DFI Interface Signals (continued)

Signal Name	I/O	Description
		The bit width of this signal is determined by the number of ranks. Taking a 2-rank design as an example, the mapped relationship is as follow: **For DDR23:** [0] is mapped to RANK0 for P0. [1] is mapped to RANK1 for P0. [2] is mapped to RANK0 for P1. [3] is mapped to RANK1 for P1. **For DDR4:** [0] is mapped to RANK0 for P0. [1] is mapped to RANK1 for P0. [2] is mapped to RANK0 for P1. [3] is mapped to RANK1 for P1. [4] is mapped to RANK2 for P0. [5] is mapped to RANK3 for P0. [6] is mapped to RANK2 for P1. [7] is mapped to RANK3 for P1. **For LPDDR4/4X:** These signals define the CAT ODT information that is intended for the LPDDR4/4X DRAM devices. **Voltage Range**: 0 V–VDD **Active State**: High **Synchronous to**: core_clk **Registered**: Yes ***Note:*** *The n represents the number of ranks.*
dfi_bank[5:0]	O	DFI bank bus.

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Table 4. DFI Interface Signals (continued)

Signal Name	I/O	Description
		Control signal defining the bank of a command. This signal is only for DDRn. For LPDDRn, these signals must be held in the idle state. **Voltage Range**: 0 V–VDD **Active State**: High **Synchronous to**: core_clk **Registered**: Yes
dfi_bg[3:0]	O	DFI bank group bus. Control signal defining the bank group of a command. **This signal is only for DDR4.** **Voltage Range**: 0 V–VDD **Active State**: High **Synchronous to**: core_clk **Registered**: Yes
dfi_reset_n[1:0]	O	DFI reset bus. These signals define the RESET information that is intended for the DRAM memory devices for all control commands. **Voltage Range**: 0 V–VDD **Active State**: Low **Synchronous to**: core_clk **Registered**: Yes
dfi_act_n[1:0]	O	DFI row active bus for DDR4. Control signal asserted for DRAM ACT commands. **This signal is only for DDR4.**

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Table 4. DFI Interface Signals (continued)

Signal Name	I/O	Description
		Voltage Range: 0 V–VDD **Active State**: N/A **Synchronous to**: core_clk **Registered**: Yes
dfi_dram_clk_disable	O	DRAM CK disable. When active, this indicates to the PHY that the CK to the DRAM devices must be disabled such that the clock signals hold a constant value. **Voltage Range**: 0 V–VDD **Active State**: High **Synchronous to**: core_clk **Registered**: Yes
DFI Low Power Interface		
dfi_lp_req	O	Low power opportunity request. This signal is used by the Memory Controller to inform the PHY of an opportunity to switch to a low power mode. **Voltage Range**: 0 V–VDD **Active State**: High **Synchronous to**: core_clk **Registered**: Yes
dfi_lp_wakeup[3:0]	O	Low power wakeup time. These signals indicate which one of the 16 wakeup times the Memory Controller is requesting for the PHY. **0000:** t_{lp_wakeup} is 16 cycles **0001:** t_{lp_wakeup} is 32 cycles **0010:** t_{lp_wakeup} is 64 cycles

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Table 4. DFI Interface Signals (continued)

Signal Name	I/O	Description
		… **1110:** t_{lp_wakeup} is 262144 cycles **1111:** t_{lp_wakeup} is unlimited **Voltage Range**: 0 V–VDD **Active State**: N/A **Synchronous to**: core_clk **Registered**: Yes
dfi_lp_ack	I	Low power acknowledge. These signals are asserted to acknowledge the Memory Controller low power opportunity request. The PHY is not required to acknowledge this request. T_{lp_resp} = 7 **Voltage Range**: 0 V–VDD **Active State**: High **Synchronous to**: core_clk **Registered**: Yes
DFI Write Data Interface		
dfi_wrdata_en[*n*/4-1:0]	O	Write data and data mask enable. **Voltage Range**: 0 V–VDD **Active State**: High **Synchronous to**: core_clk **Registered**: Yes ***Note:*** *The n represents the number of DQs.*
dfi_wrdata[4*n*-1:0]	O	Write data bus.

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Table 4. DFI Interface Signals (continued)

Signal Name	I/O	Description
		Voltage Range: 0 V–VDD **Active State**: N/A **Synchronous to**: core_clk **Registered**: Yes ***Note:*** *The n represents the number of DQs.*
dfi_wrdata_mask[*n*/2-1:0]	O	Write data mask bus. **Voltage Range**: 0 V–VDD **Active State**: High **Synchronous to**: core_clk **Registered**: Yes ***Note:*** *The n represents the number of DQs.*
DFI Read Data Interface		
dfi_rddata_en[*n*/4-1:0]	O	Read data enable. **Voltage Range**: 0 V–VDD **Active State**: High **Synchronous to**: core_clk **Registered**: Yes ***Note:*** *The n represents the number of DQs.*
dfi_rddata[4*n*-1:0]	I	Read data bus. **Voltage Range**: 0 V–VDD **Active State**: N/A **Synchronous to**: core_clk

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Table 4. DFI Interface Signals (continued)

Signal Name	I/O	Description
		Registered: Yes ***Note:*** *The n represents the number of DQs.*
dfi_rddata_dbi[*n*/2-1:0]	I	The read data bus inversion (byte-based). When the signal is low, The corresponding read data need inversion. **Voltage Range**: 0 V–VDD **Active State**: High **Synchronous to**: core_clk **Registered**: Yes ***Note:*** *The n represents the number of DQs.*
dfi_rddata_valid*	I	Read data valid indicator. **Voltage Range**: 0 V–VDD **Active State**: High **Synchronous to**: core_clk **Registered**: Yes
DFI Status Interface		
dfi_init_start	O	The PHY initialization start signal. **Voltage Range**: 0 V–VDD **Active State**: Low **Synchronous to**: core_clk
dfi_frequency[4:0]	O	The PHY initialization frequency point select signal. **5'h0:** Frequency Point 0 **5'h1:** Frequency Point 1 **5'h2:** Frequency Point 2

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Table 4. DFI Interface Signals (continued)

Signal Name	I/O	Description
		5'h3: Frequency Point 3 **Voltage Range**: 0 V–VDD **Active State**: N/A **Synchronous to**: core_clk **Registered**: Yes
dfi_init_complete	I	The PHY initialization complete signal. **Voltage Range**: 0 V–VDD **Active State**: High **Synchronous to**: core_clk **Registered**: Yes
DFI Controller Update Interface		
dfi_ctrlupd_req	O	Controller update request signal from the controller. **Active State**: High **Synchronous to**: core_clk
dfi_ctrlupd_ack	I	Controller update acknowledge signal from the PHY. **Active State**: High **Synchronous to**: core_clk
dfi_ctrlupd_req_type[1:0]	O	Controller update request type signal from the controller. **2'b00:** Type 0 DFI Control Update. **2'b01:** Type 1 DFI Control Update. **2'b10:** Type 2 DFI Control Update. **2'b11:** Unused. **Synchronous to**: core_clk

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Table 4. DFI Interface Signals (continued)

Signal Name	I/O	Description
DFI PHY Master Interface		
dfi_phymstr_ack	O	DFI PHY master acknowledge. When asserted, the MC places the DRAM in a known state (IDLE, self-refresh, or self-refresh power down). When the **dfi_phymstr_ack** signal is asserted, the PHY is the master of DRAM bus. If required by the DRAM, the controller continues sending refresh commands on the DFI bus. **Active State**: High **Synchronous to**: core_clk
dfi_phymstr_req	I	DFI PHY master request. When asserted, the PHY requests control of the DFI bus. **Active State**: High **Synchronous to**: core_clk

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3.4. APB Slave Interface Signals

Table 5. APB Slave Interface Signals

Signal Name	I/O	Description
apb_pclk	I	APB clock The rising edge is used to time all transfer on the APB.
apb_presetn	I	Reset This signal is normally connected directly to the system bus reset signal. **Active State**: Low **Synchronous to**: apb_pclk
apb_paddr[31:0]	I	APB address bus **Synchronous to**: apb_pclk
apb_psel	I	Select Indicates that the slave device is selected and a data transfer is required. **Active State**: High **Synchronous to**: apb_pclk
apb_penable	I	Enable Indicates the second cycle of an APB transfer. **Active State**: High **Synchronous to**: apb_pclk
apb_pwrite	I	Direction Indicates an APB write access when HIGH and an APB read access when LOW. **Synchronous to**: apb_pclk

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Table 5. APB Slave Interface Signals (continued)

Signal Name	I/O	Description
apb_prdata[31:0]	O	Read data The selected slave drives this bus during read cycles when **apb_pwrite** is low. **Synchronous to**: apb_pclk
apb_pwdata[31:0]	I	Write data This bus is driven by the peripheral bus bridge unit during write cycles when **apb_pwrite** is high. **Synchronous to**: apb_pclk
apb_pready	O	Ready The slave uses this signal to extend an APB transfer. **Active State**: High **Synchronous to**: apb_pclk
apb_pstrb[*n*-1:0]	I	Write strobes When asserted HIGH, a write strobe indicates that the corresponding byte lane of the write data bus contains valid information. This is an APB 4.0 signal. ***Note:*** *The n represents the number of bytes.*

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3.5. DFT Scan Interface Signals

Table 6. DFT Scan Interface

Signal Name	I/O	Description
imc_scan_rstn	I	Reset of scan path **Active State**: Low
imc_scan_mode	I	Mode select signal of scan **Active State:** High

3.6. Interrupt Interface

Table 7. Interrupt Interface

Signal Name	I/O	Description
imc_interrupt	O	Interrupts for controller **Active State:** High **Synchronous to:** core_clk

Table 8. Interrupt Type

Interrupt Type	Description
waddr_in_out_of_range_to_int	When the AXI write access address exceeds the SDRAM Density setting by the reg_rank/bank/row/col_size and reg_sdram_dq_width, this signal will change to high and keep until the corresponding interrupt **clear** signa sets to 1'b1. If the user wants to mask this interrupt, the user can set the corresponding **mask** signal to 1'b1. The SDRAM Density = 2 ^ (reg_rank_size + reg_bank_size + reg_row_size + reg_col_size + reg_sdram_dq_width) * 1 Byte.
raddr_in_out_of_range_to_int	When the AXI read access address exceeds the SDRAM Density setting by the reg_rank/bank/row/col_size and

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Table 8. Interrupt Type (continued)

Interrupt Type	Description
	reg_sdram_dq_width, this signal will change to high and keep until the corresponding interrupt **clear** signal sets to 1'b1. If the user wants to mask this interrupt, the user can set the corresponding **mask** signal to 1'b1. The SDRAM Density = 2 ^ (reg_rank_size + reg_bank_size + reg_row_size + reg_col_size + reg_sdram_dq_width) * 1 Byte.
link_ecc_rx_error_int	When the IMC and PHY enable the Link ECC function, if the PHY detects a Link ECC RX direction error, the DFI interface signal **dfi_rx_resp_valid** will changes to high and the **dfi_rx_resp** will changes to un-zero state, then this signal will change to high and keep until the corresponding interrupt **clear** signal sets to 1'b1. If the user wants to mask this interrupt, the user can set the corresponding **mask** signal to 1'b1. The user can through the only read register ***reg_rx_resp_error_cnt*** to know the numbers of the Link ECC RX Error.
dfi_tctrlupd_min_vio_type0_int	For the MC-Initiated Update interface, the MC must assert the **dfi_ctrlupd_req** signal for at least $\mathbf{t_{ctrlupd_min}}$ time and hope the PHY to assert the **dfi_ctrlupd_ack** during this time to complete the handshake. If the controller can't wait the assert of the **dfi_ctrlupd_ack** during this time and the **dfi_ctrlupd_req_type** is type0 (which value is 2'b00), then this signal will change to high and keep until the corresponding interrupt **clear** signal sets to 1'b1. If the user wants to mask this interrupt, the user can set the corresponding **mask** signal to 1'b1. The $\mathbf{t_{ctrlupd_min}}$ timing parameter is set by the register ***reg_ddrc_dfi_ctrlupd_min***. The suggest value is 0x10.
dfi_tctrlupd_min_vio_type1_int	For the MC-Initiated Update interface, the MC must assert the **dfi_ctrlupd_req** signal for at least $\mathbf{t_{ctrlupd_min}}$ time and hope the PHY to assert the **dfi_ctrlupd_ack** during this time to complete

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Table 8. Interrupt Type (continued)

Interrupt Type	Description
	the handshake. If the controller can't wait the assert of the **dfi_ctrlupd_ack** during this time and the **dfi_ctrlupd_req_type** is type1 (which value is 2'b01), then this signal will change to high and keep until the corresponding interrupt **clear** signal sets to 1'b1. If the user wants to mask this interrupt, the user can set the corresponding **mask** signal to 1'b1. The $\mathbf{t_{ctrlupd_min}}$ timing parameter is set by the register ***reg_ddrc_dfi_ctrlupd_min_type1***. The suggest value is 0x10.
dfi_tctrlupd_min_vio_type2_int	For the MC-Initiated Update interface, the MC must assert the **dfi_ctrlupd_req** signal for at least $\mathbf{t_{ctrlupd_min}}$ time and hope the PHY to assert the **dfi_ctrlupd_ack** during this time to complete the handshake. If the controller can't wait the assert of the **dfi_ctrlupd_ack** during this time and the **dfi_ctrlupd_req_type** is type2 (which value is 2'b10), then this signal will change to high and keep until the corresponding interrupt **clear** signal sets to 1'b1. If the user wants to mask this interrupt, the user can set the corresponding **mask** signal to 1'b1. The $\mathbf{t_{ctrlupd_min}}$ timing parameter is set by the register ***reg_ddrc_dfi_ctrlupd_min_type1***. The suggest value is 0x10.
dfi_tctrlupd_max_vio_type0_int	For the MC-Initiated Update interface, the total number of cycles that the **dfi_ctrlupd_req** signal is asserted must not exceeds $\mathbf{t_{ctrlupd_max}}$ time and hope the PHY can complete the parameter update or Re-Training during the $\mathbf{t_{ctrlupd_max}}$. If the PHY can't complete the parameter update or Re-Training during the $\mathbf{t_{ctrlupd_max}}$ to de-assert of the **dfi_ctrlupd_ack** during this time and the **dfi_ctrlupd_req_type** is type0 (which value is 2'b00), then this signal will change to high and keep until the corresponding interrupt **clear** signal sets to 1'b1. If the user wants to mask this interrupt, the user can set the corresponding **mask** signal to 1'b1.

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Table 8. Interrupt Type (continued)

Interrupt Type	Description
	The $\mathbf{t_{ctrlupd_max}}$ timing parameter is set by the register ***reg_ddrc_dfi_ctrlupd_max***. The suggest value is 0x20.
dfi_tctrlupd_max_vio_type1_int	For the MC-Initiated Update interface, the total number of cycles that the **dfi_ctrlupd_req** signal is asserted must not exceeds $\mathbf{t_{ctrlupd_max}}$ time and hope the PHY can complete the parameter update or Re-Training during the $\mathbf{t_{ctrlupd_max}}$. If the PHY can't complete the parameter update or Re-Training during the $\mathbf{t_{ctrlupd_max}}$ to de-assert of the **dfi_ctrlupd_ack** during this time and the **dfi_ctrlupd_req_type** is type1 (which value is 2'b01), then this signal will change to high and keep until the corresponding interrupt **clear** signal sets to 1'b1. If the user wants to mask this interrupt, the user can set the corresponding **mask** signal to 1'b1. The $\mathbf{t_{ctrlupd_max}}$ timing parameter is set by the register ***reg_ddrc_dfi_ctrlupd_max_type1***. The suggest value is 0x200.
dfi_tctrlupd_max_vio_type2_int	For the MC-Initiated Update interface, the total number of cycles that the **dfi_ctrlupd_req** signal is asserted must not exceed $\mathbf{t_{ctrlupd_max}}$ time and hope the PHY can complete the parameter update or Re-Training during the $\mathbf{t_{ctrlupd_max}}$. If the PHY can't complete the parameter update or Re-Training during the $\mathbf{t_{ctrlupd_max}}$ to de-assert of the **dfi_ctrlupd_ack** during this time and the **dfi_ctrlupd_req_type** is type2 (which value is 2'b10), then this signal will change to high and keep until the corresponding interrupt **clear** signal sets to 1'b1. If the user wants to mask this interrupt, the user can set the corresponding **mask** signal to 1'b1. The $\mathbf{t_{ctrlupd_max}}$ timing parameter is set by the register ***reg_ddrc_dfi_ctrlupd_max_type1***. The suggest value is 0x200.
dfi_tphymstr_resp_vio_int	For the PHY-Initiated Update interface, the **dfi_phyupd_ack** signal must assert within $\mathbf{t_{phyupd_resp}}$ cycles after the assertion of the **dfi_phyupd_req** signal. If the IMC can't assert the **dfi_phyupd_ack** during the $\mathbf{t_{phyupd_resp}}$, then this signal will

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Table 8. Interrupt Type (continued)

Interrupt Type	Description
	change to high and keep until the corresponding interrupt **clear** signal sets to 1'b1. If the user wants to mask this interrupt, the user can set the corresponding **mask** signal to 1'b1. The tphyupd_resp timing parameter is set by the register ***reg_ddrc_tphymstr_resp***. It depends on the time that the IMC assert the **dfi_phyupd_ack** after the assert of the **dfi_phyupd_req.** The suggest value is 50ns.
dfi_tphymstr_type_vio_int	For the PHY-Initiated Update interface, when the **dfi_phyupd_ack** signal is asserted, it must remain asserted until the **dfi_phyupd_req** signal de-asserts. If the **dfi_phyupd_req** signal can't de-asserts during that time , then this signal will change to high and keep until the corresponding interrupt **clear** signal sets to 1'b1. If the user wants to mask this interrupt, the user can set the corresponding **mask** signal to 1'b1. The $\mathbf{t_{phyupd_type}}$ timing parameter is set by the register **reg_ddrc_tphymstr_resp**. The suggest value is 500ns.

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3.7. SRAM Interface

Table 9. SRAM Interface

Signal Name	I/O	Description
imc_sram_*q*	I	The read data of SRAM in controller.
imc_sram_*d*	O	The write data of SRAM in controller.
imc_sram_*wen*	O	The write enable signal of SRAM in controller. **Active State:** Low
imc_sram_*ren*	O	The read enable signal of SRAM in controller. **Active State:** Low
imc_sram_*waddr*	O	The write address of SRAM in controller.
imc_sram_*raddr*	O	The read address of SRAM in controller.

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4

Timing Specifications

4.1. APB Bus Timing

4.1.1. System Configure Write Timing for APB Bus

The write transfer starts with the address, write data, write signal and select signal all changing after the rising edge of the clock. The first clock cycle of the transfer is called the Setup phase. After the first clock edge, the enable signal **apb_penable** is asserted, which indicates that the Access phase is taking place. The address, data and control signals all remain valid throughout the Access phase. The transfer completes at the end of this cycle.

The enable signal , **apb_penable**, is de-asserted at the end of the transfer. The select signal, **apb_psel**, also becomes LOW unless another transfer to the same peripheral follows immediately.

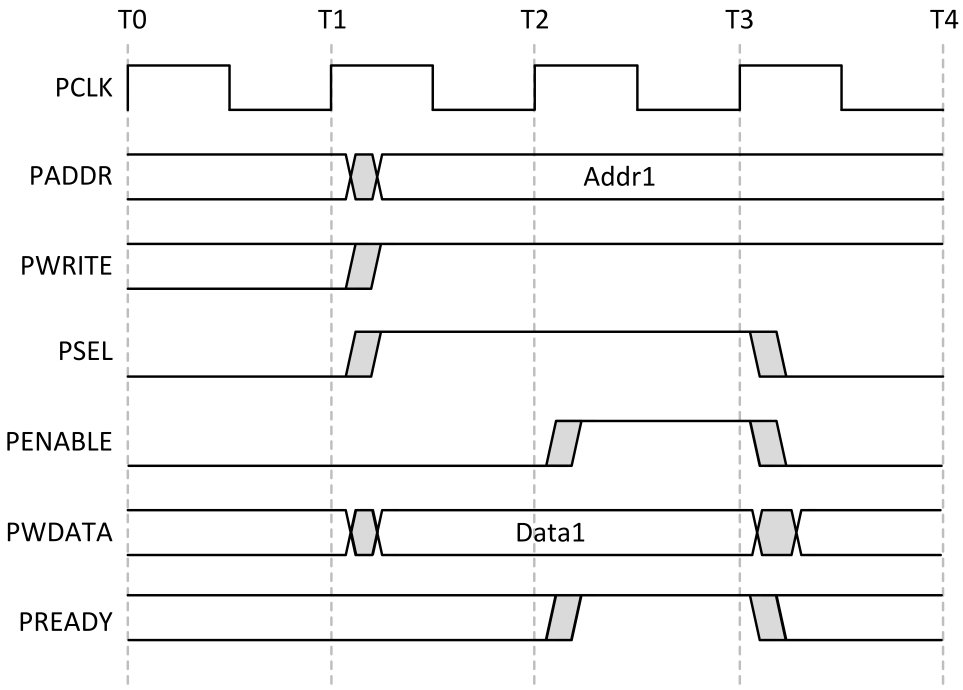

Figure 6. APB Write Transfer Without Wait States

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Figure 6 (on page 51) shows how the **apb_pready** signal from the slave can extend the transfer. During an Access phase, when **apb_penable** is HIGH, the transfer can be extended by driving **apb_pready** LOW. The following signals remain unchanged for the additional cycles:

- Address, **apb_paddr**
- Write signal, **apb_pwrite**
- Select signal, **apb_psel**
- Enable signal, **apb_penable**
- Write data, **apb_pwdata**

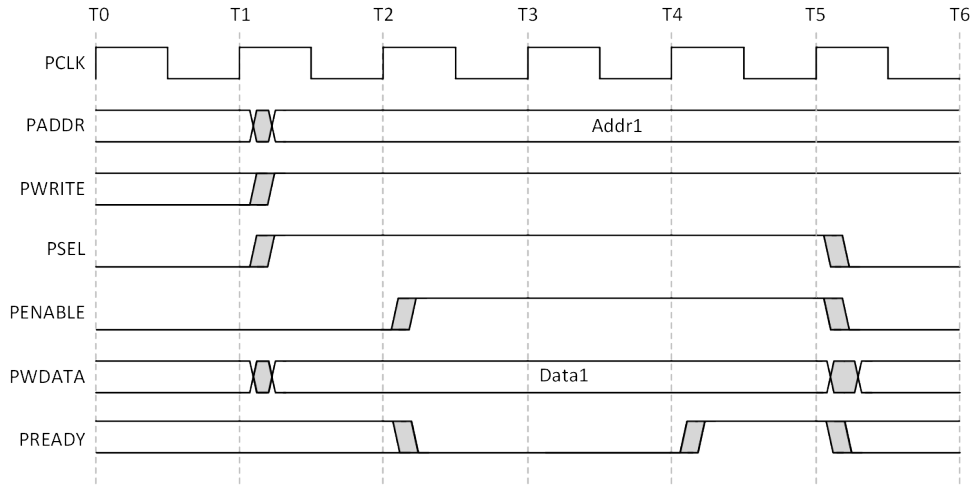

Figure 7. APB Write Transfer With Wait States

apb_pready can take any values when **apb_penable** is LOW. This ensures that peripherals with a fixed two-cycle access can tie **apb_pready** to HIGH.

4.1.2. System Configure Read Timing for APB Bus

Figure 8 (on page 53) shows a read transfer. The timing of the address, write, select, and enable signals are as described in Write transfers. The slave must provide the data before the end of the read transfer.

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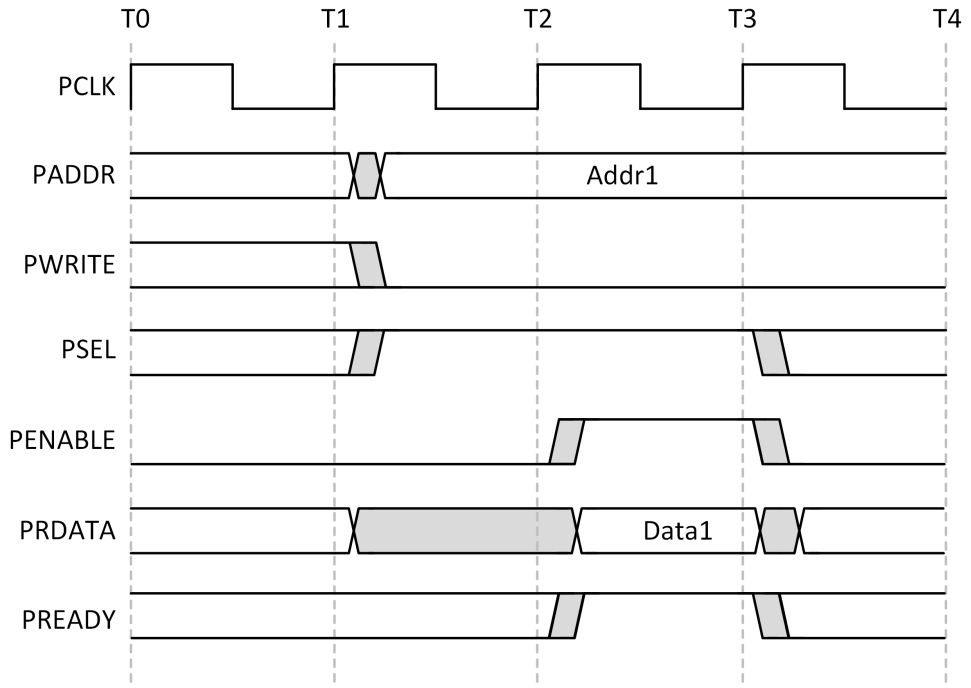

Figure 8. APB Read Transfer without Wait States

Figure 8 (on page 53) shows how the **apb_pready** signal can extend the transfer. The transfer is extended if **apb_pready** is driven LOW during an Access phase. The protocol ensures that the following remain unchanged for the additional cycles:

- Address, **apb_paddr**
- Read signal, **apb_pwrite**
- Select signal, **apb_psel**
- Enable signal, **apb_penable**

Figure 9 (on page 53) shows two cycles are added using the **apb_pready** signal. However, you can add as many additional cycles as possible, starting from zero.

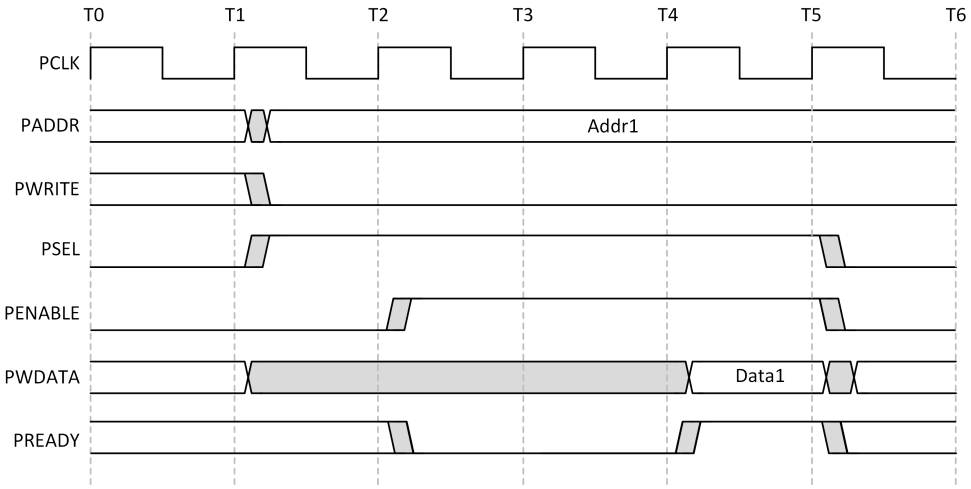

Figure 9. APB Read Transfer with All States

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4.2. AXI Bus Timing

The AXI protocol is burst-based. Every transaction has address and control information on the address channel, describing the nature of the data to be transferred. The data is transferred using a write data channel to the slave or a read data channel to the master.

4.2.1. AXI Write Transaction Timing

The write data channel conveys the write data from the master to the slave and includes:

- The data bus, referring to the signal wdata_*[2^n-1:0] which can be 2^3, 2^4, 2^5, ..., 2^n bits wide bits wide.
- One byte lane strobe for every data bit, indicating which bytes of the data bus are valid.

In write transactions, data flowing from the master to the slave, the AXI protocol has an additional write response channel to allow the slave to send signal to the master, informing the completion of the write transaction. *Figure 10 (on page 54)* gives an example timing of the write burst.

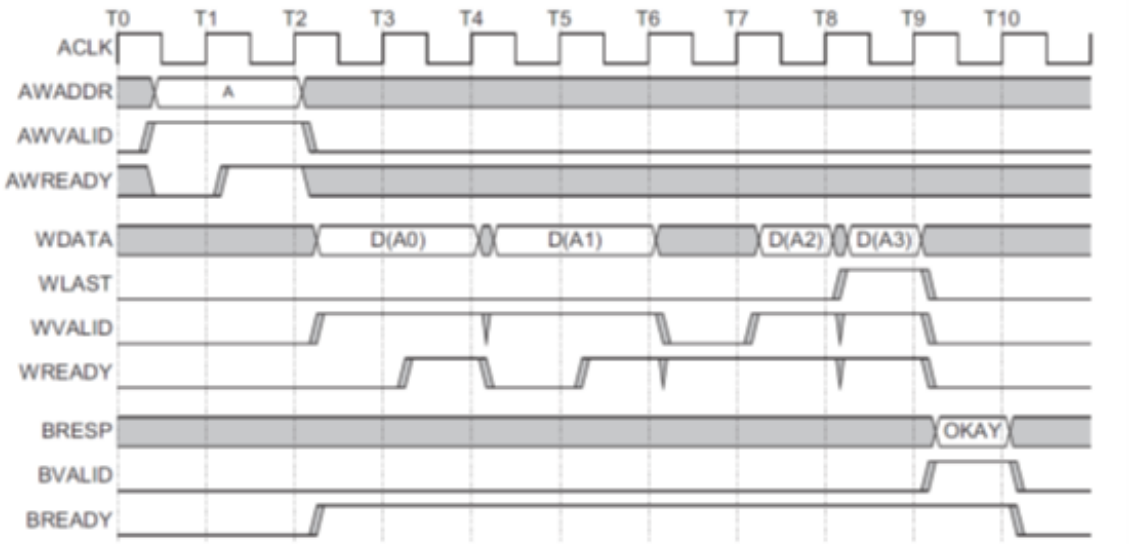

Figure 10. Write Burst

4.2.2. AXI Read Transaction Timing

The read data channel conveys both the read data and any read response information from the slave back to the master. The read data channel includes:

- The data bus, referring to the signal rdata_*[2^n-1:0] which can be 2^3, 2^4, 2^5, ..., 2^n bits wide bits wide.
- A read response indicating the completion status of the read transaction.

Figure 11 (on page 55) shows a read burst of four transfers.

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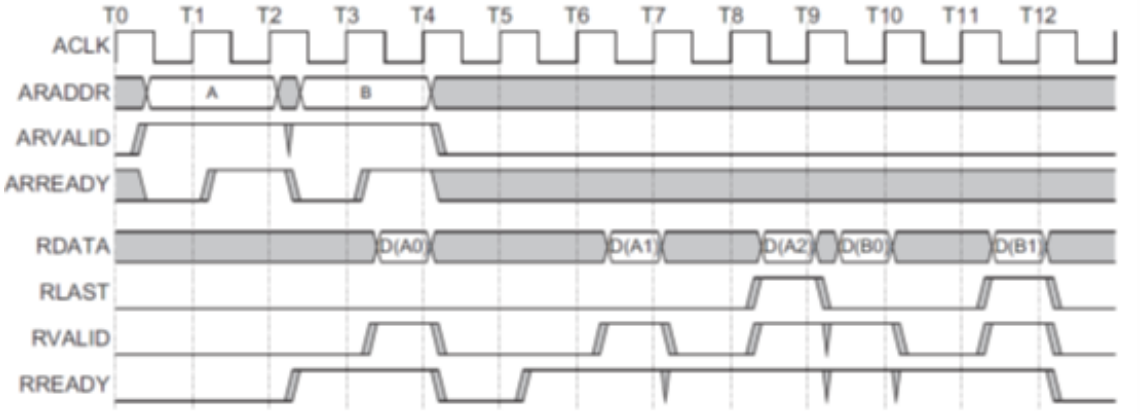

Figure 11. Read Burst

All five AXI channels use the same **VALID**/**READY** handshake to transfer data and control information. This two-way flow control mechanism enables both the master and the slave to control the rate at which the data and control information are transferred. The source generates the **VALID** signal to indicate when the data or control information is available. The destination generates the **READY** signal to indicate that it is ready to accept the data or control information. Transfer occurs only when both the **VALID** and **READY** signals are HIGH.

In following figure, the source presents the data or control information and drives the **VALID** signal HIGH. The data or control information from the source remains stable until the destination drives the **READY** signal HIGH, indicating that it is ready to accept the data or control information. The arrow indicates the time when the transfer occurs.

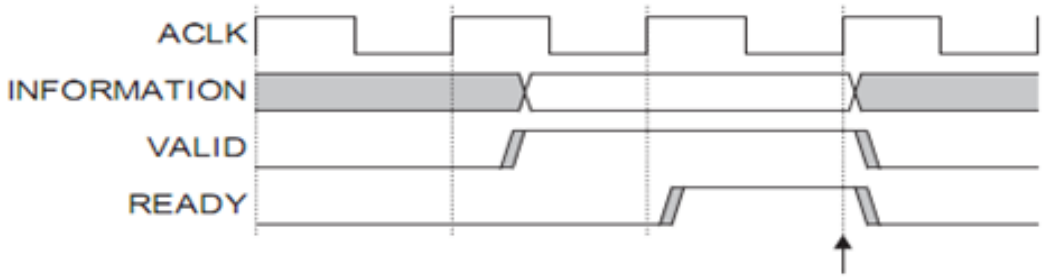

Figure 12. Valid Before Ready Handshake

In the following *Figure 13 (on page 56)*, the destination drives **READY** HIGH before the data or control information is valid. This indicates that the destination can accept the data or control information in a single cycle as soon as it becomes valid. The arrow indicates the time when transfer occurs.

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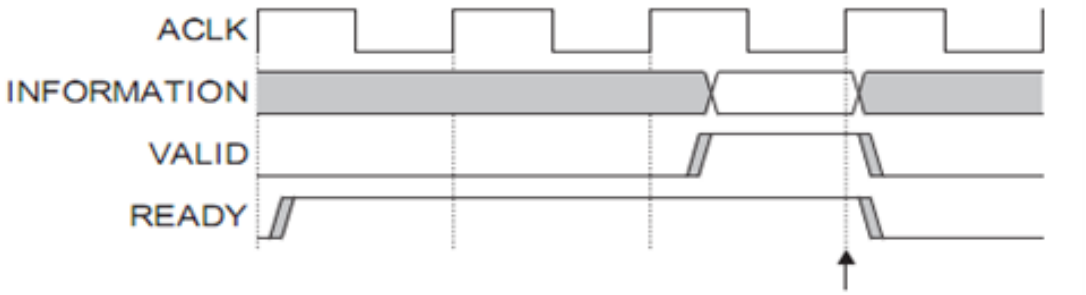

Figure 13. Ready Before Valid Handshake

Figure 14 (on page 56) shows when the source and destination occur in the same cycle, they can transfer data or control information. In this case transfer occurs immediately. The arrow indicates the time when the transfer occurs.

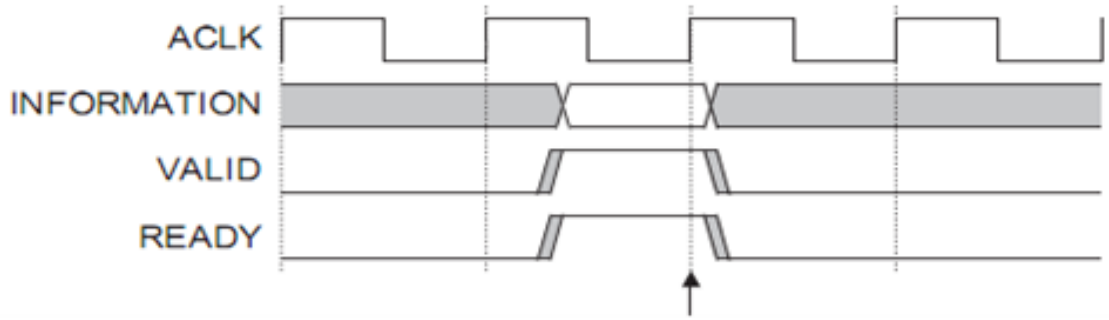

Figure 14. Ready with Valid Handshake

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4.3. DFI Timing

4.3.1. DFI Write Data Interface Timing

The definition of timing parameters related to the write data interface are in the databook. When the IDC connects to the INNO DDR2/3/3L/4/LPDDR2/3/4/4X PHY, the recommended values are shown in *Table 10 (on page 57)*.

Table 10. DFI Write Timing Parameters

SDRAM	Parameter	Unit	Value	Description
DDR2 DDR3/3L DDR4	t_{phy_wrdata}	core_clock cycles	0	Specifies the number of core clock cycles between when the **dfi_wrdata_en** signal is asserted and when the associated write data is driven on the **dfi_wrdata** signal.
	t_{phy_wrlat}	core_clock cycles	((CWL+AL-1)>>1)+2	Specifies the number of core clock cycles between when a write command is sent on the DFI control interface and when the **dfi_wrdata_en** signal is asserted.
LPDDR2 LPDDR3 LPDDR4/4X	t_{phy_wrdata}	core_clock cycles	0	Same as t_{phy_wrdata} above.
	t_{phy_wrlat}	core_clock cycles	(CWL>>1)+2	Same as t_{phy_wrlat} above.

The t_{phy_wrdata} is 0. So, for IDC, the **dfi_wrdata** is aligned to the **dfi_wrdata_en**. And the t_{phy_wrlat} defines the delay between the DFI write command to the **dfi_wrdata** and **dfi_wrdata_en**.

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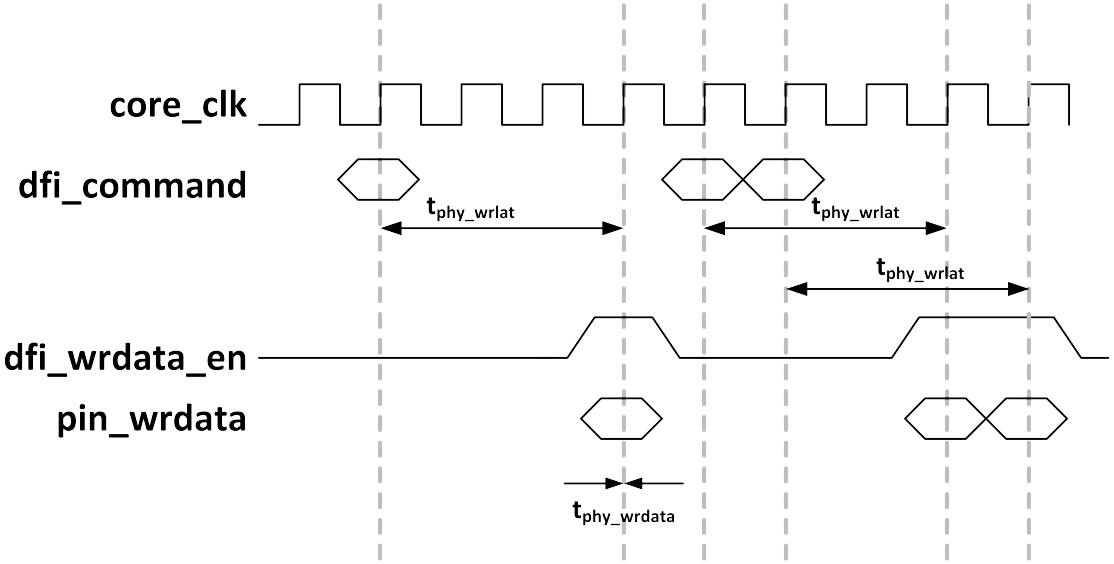

Figure 15. Write Timing of DFI Specification

4.3.2. DFI Read Data Interface Timing

The definition of timing parameters related to the read data interface are in the databook. When the IDC connects to the INNO DDR2/3/3L/4/LPDDR2/3/4/4X PHY, the recommended values are shown in *Table 11 (on page 58)*.

Table 11. DFI Read Timing Parameters

SDRAM	Parameter	Unit	Value	Description
DDR2 DDR3/3L DDR4	**t_{rdata_en}**	core_clock cycles	((CL+AL-1)>>1)+1	Specifies the number of core clock cycles from the assertion of a read command on the DFI to the assertion of the **dfi_rddata_en** signal.
LPDDR2 LPDDR3	**t_{rdata_en}**	core_clock cycles	(CL>>1)+1	Same as above
LPDDR4/4X	t_{rdata_en}	core_clock cycles	(CL>>1)+2	Same as above

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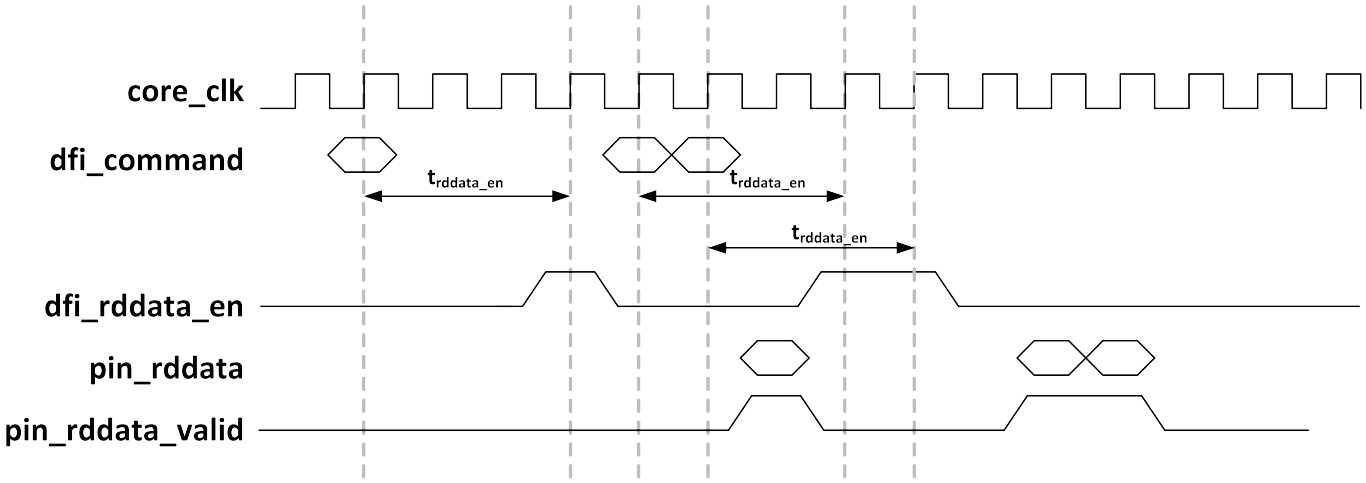

Figure 16. Read Timing of DFI Specification

4.4. Reset Timing

The Controller's requirements for reset timing are as follows:

Step 1 After system is powered on, set all the reset signals to low to achieve the reset state of all modules. This state lasts for at least two clock cycles in respective modules time domains.

Step 2 Provide stable clock input to all clock pins.

Step 3 Release the reset signal of APB BUS, and configure the register through the APB BUS. During this time, both AXI module and the CORE module are in the reset state. Therefore, there is no metastable state caused by cross domain issues during register configuration.

Step 4 After completing the configuration of all registers (these registers are static, and there is no two-stage synchronization in the design), release the CORE module reset signal.

Step 5 After the CORE reset is released, wait for at least two clock cycles to complete the initialization of PHY and Controller, and complete the training of PHY and SDRAM.

Step 6 Release the reset of the AXI BUS, and wait for at least two clock cycles to start read and write operations.

Figure 17 (on page 60) shows the Controller reset timing

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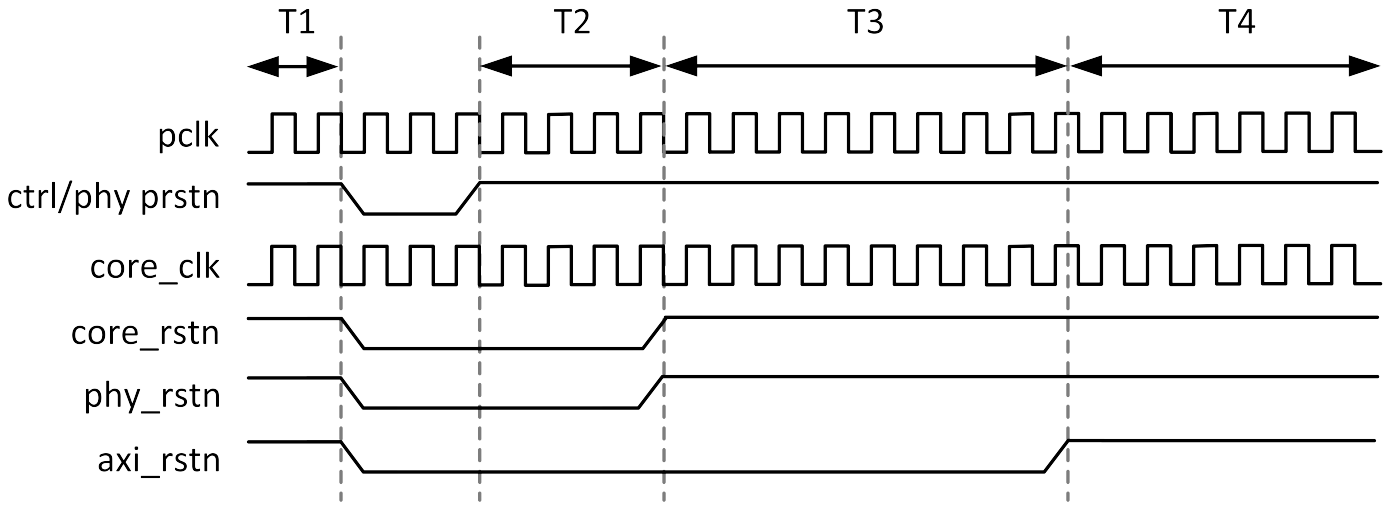

Figure 17. Controller Reset Timing

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5

Deliverables

No.	Description
1	Controller User Guide of IP Pacakage
2	Controller Databook
3	Controller Integration Guide
4	Controller Application Guide
5	Controller User Guide of UVM Simulation Environment
6	Controller Checklist of Verification
7	Controller Verification Report (Optional)
8	Controller nLint/CDC Report (Optioanl)
9	Controller User Guide of Hardware Emulator Model (Optional)
10	Controller Register Table, RALF Model or XML Formalt Register File
11	Controller Zebu/Palladium Model pulse DFI Interface RAM Module (Optional)
12	Controller Latency Analysis Tool (Optional)
13	Controller No-timing checklist
14	Controller Synthesis Constraint File
15	Controller Scramble RTL

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